Jail Overcrowding by Region and Size, Luke Dougherty and Jony Dobkin

https://github.com/jonydobkin6/STAT-project-Jails-Jony-Dobkin-Luke-Dougherty/tree/main

Jail Dataset After Modifications

```
##                                NAME TOTPOP RATED STATE region jail_size
## 1     Baldwin County Sheriffs Office    479   654     1  South     Large
## 2    Cherokee County Sheriffs Office    134   140     1  South     Small
## 3 Chilton County Sheriffs Department    232   232     1  South     Small
## 4  Clarke County Sheriffs Department    115   120     1  South     Small
## 5     Cullman County Sheriffs Office    428   553     1  South    Medium
## 6    Escambia County Sheriffs Office    285   132     1  South     Small
##   crowding_rate
## 1         73.24
## 2         95.71
## 3        100.00
## 4         95.83
## 5         77.40
## 6        215.91
```

Introduction

Jail overcrowding has become one of the most pressing challenges in the United States today. When a jail's population exceeds its capacity, facilities face greater risks of violence, limited access to medical care, and increased strain on staff. Understanding where jail overcrowding is worst is important for policymakers trying to address these issues. This project examines two questions: whether certain geographic regions experience higher mean jail occupancy

rates than others and whether smaller jails have higher mean occupancy rates than larger ones. By analyzing these factors, we can better identify where the jail system is under the most pressure and focus attention on those areas.

Data Summary

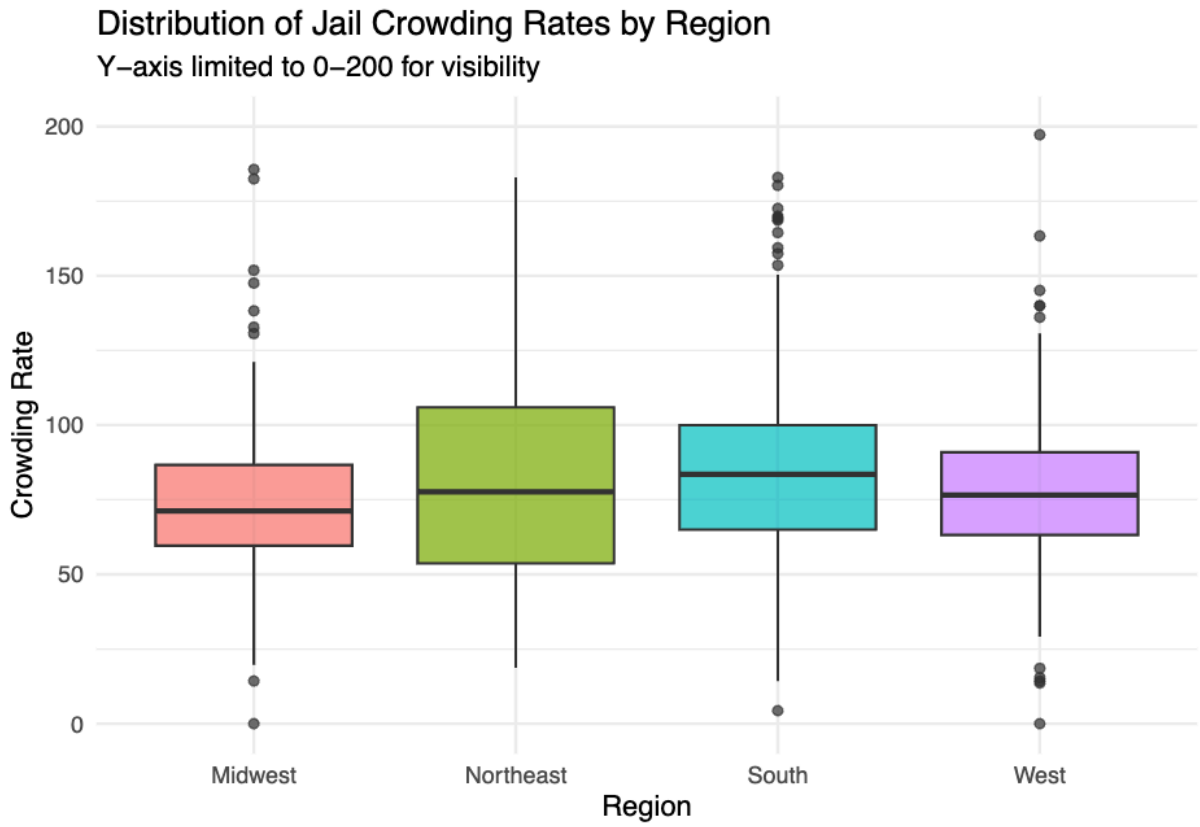
This project's dataset comes from the 2023 Annual Survey of Jails (ASJ), published by the U.S. Bureau of Justice Statistics (BJS), which collects records directly from county and municipal jail facilities across the United States. Rather than a full census, the ASJ uses a stratified probability sampling design to select about 950 jails representing the approximately 2,850 jails nationwide. Facilities are grouped into ten strata based on inmate population and the presence of juveniles, then randomly selected from those strata (two strata of larger jails ended up having all of their jails selected). BJS collects this data annually to monitor jail operations, assess crowding, and inform policymakers about trends in local corrections.

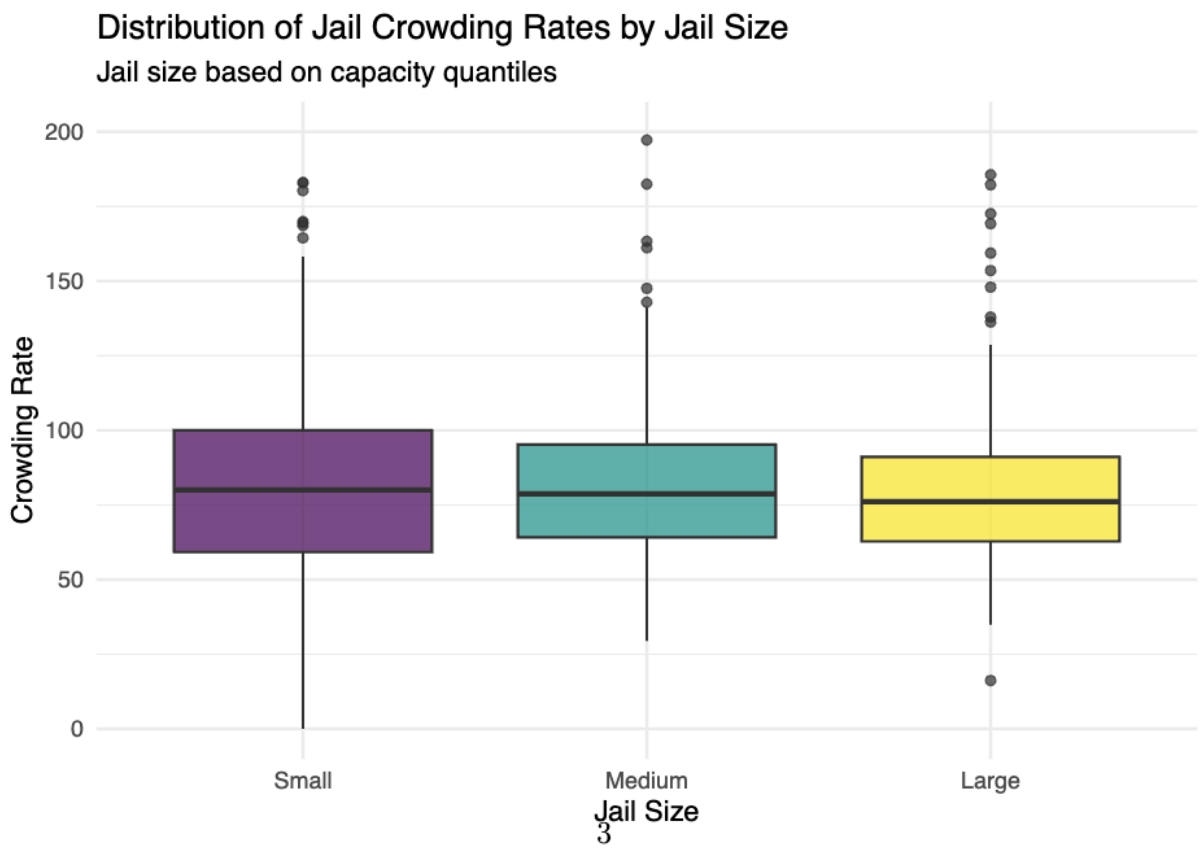
Several modifications were made after loading the data. First, we subsetted for only the most important variables, facility name, state, total population, and rated capacity, which made the data easier to work with, given the original 150 columns. We then created our dependent variable, a crowding rate, by dividing the total population by rated capacity and multiplying by 100. Next, we created two independent variables: region, categorizing jails into four groups (Northeast, South, Midwest, and West), and size (Small, Medium, Large) based on rated capacity. Finally, we removed extreme outliers to eliminate likely data entry errors. One row, for example, showed a 600% crowding rate, which subsequent research confirmed was erroneous.

There are several potential issues with the data and our modifications. While large jails hold the majority of the inmate population, most jail jurisdictions in the United States are small or rural facilities. Because the ASJ samples only a fraction of these smaller jurisdictions while including all large ones, findings for "Small" and "Medium" facilities may be less precise than those for "Large" facilities. Additionally, rated capacity is not standardized across the country, and facilities may define it differently, which could make comparisons across groups less accurate. Finally, jails with missing capacity values were excluded from crowding rate calculations, which could introduce bias if missingness correlates with facility size or region.

Despite these limitations, we believe this dataset is appropriate for the research question. It covers a broad, nationally representative sample of jail facilities, and its population and capacity measures are the most consistent available, making it well-suited for regional and size-based comparisons even if imperfect.

Exploratory Data Analysis





```
## # A tibble: 4 x 2
##   region    avg_crowding_rate
##   <chr>                 <dbl>
## 1 Midwest                74.3
## 2 Northeast              84.4
## 3 South                  86.8
## 4 West                   77.7
```

```
## # A tibble: 3 x 2
##   jail_size avg_crowding_rate
##   <ord>                 <dbl>
## 1 Small                  82.6
## 2 Medium                 81.8
## 3 Large                  79.0
```

Conclusions

There were many interesting conclusions to draw from the data. Southern jails have the highest average crowding rate at 86.8%, followed by the Northeast at 84.4%, the West at 77.7%, and the Midwest at 74.3%, marking a 12-point gap between the highest and lowest regions. The boxplots also reveal differences in spread and shape across regions. The Midwest displays the most compact range, with the middle 50% of facilities clustered around the median, suggesting that most Midwestern jails operate at moderate occupancy with limited variation. The Northeast, by contrast, shows the widest interquartile range of any region, indicating the most variability in crowding. The South and West fall in between, though both show numerous high outliers, with several facilities operating above 150% of rated capacity. The presence of extremely high-end outliers in almost every region confirms that dangerously overcrowded jails are not isolated to any one part of the country, though they appear most frequently in the South.

Differences by facility size are more modest. Small jails average 82.6% capacity, Medium jails 81.8%, and Large jails 79.0%. Small jails display the widest overall range, with the lower whisker extending near zero and outliers reaching well above 150%. This makes sense given how quickly they can fill up. Medium jails show a similar but slightly narrower pattern, while Large jails have the most compact distribution, with a tighter interquartile range and fewer extreme outliers. This suggests larger facilities tend to operate at more consistent occupancy levels, with less dramatic swings in either direction.

Overall, these findings show that geography is more strongly associated with crowding than facility size, though both variables reveal important distributional differences worth examining. No region or size category is immune to extreme overcrowding; high outliers appear universally, but the South has both the most outliers and the highest average crowding rate, making it the clearest target for reform.

References

Data Source: Bureau of Justice Statistics. (2023). Annual Survey of Jails, 2023 (ICPSR 39202) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/ICPSR39202.v1

Prison overcrowding. Penal Reform International. (2021, May 20). http://penalreform.org/issues/prison-conditions/key-facts/overcrowding/